

Statistical Analysis of Diabetes Type-II

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```
library(conflicted)
library(ggplot2)
library(dplyr)
library(pacman)
library(janitor)
library(skimr)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0      v stringr   1.5.1
## v lubridate 1.9.3      v tibble   3.2.1
## v purrr     1.0.2      v tidyr    1.3.0
## v readr     2.1.4
```

```
library(ADGofTest)

set.seed(1000)
```

1. Check you working directory

```
getwd()
```

```
## [1] "C:/Users/USER/Documents/Ralu Anly 506"
```

2. Set your working directory to file path

```
setwd('C:/Users/USER/Documents/Ralu Anly 506')
print(getwd())
```

```
## [1] "C:/Users/USER/Documents/Ralu Anly 506"
```

3. Open file

```
diabetes_df <- read.csv('diabetes_dataset__2019.csv', header = TRUE)

# from the janitor library, the clean_names function ensures variable names consistent
diabetes_df <- diabetes_df |> clean_names()

# check data for comprehensive summary and glimpse
skim(diabetes_df)
```

Table 1: Data summary

Name	diabetes_df
Number of rows	952
Number of columns	18
Column type frequency:	
character	14
numeric	4
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
age	0	1	5	12	0	4	0
gender	0	1	4	6	0	2	0
family_diabetes	0	1	2	3	0	2	0
high_bp	0	1	2	3	0	2	0
physically_active	0	1	4	20	0	4	0
smoking	0	1	2	3	0	2	0
alcohol	0	1	2	3	0	2	0
regular_medicine	0	1	1	3	0	3	0
junk_food	0	1	5	12	0	4	0
stress	0	1	6	10	0	4	0
bp_level	0	1	3	7	0	6	0
pdiabetes	0	1	0	3	1	4	0
urination_freq	0	1	8	11	0	2	0
diabetic	0	1	0	3	1	4	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
bmi	4	1.00	25.76	5.40	15	22	25	29	45	
sleep	0	1.00	6.95	1.27	4	6	7	8	11	
sound_sleep	0	1.00	5.50	1.87	0	4	6	7	11	
pregnancies	42	0.96	0.39	0.91	0	0	0	0	4	

4. Data Preprocessing

- Check for missing values

Summary:

Some missing values are in the ‘bmi’ and ‘pregnancies’ columns. Since ‘pregnancies’ only have one value, we’ll take the median of the values we have to fill in the gaps. For ‘bmi,’ we’ll first identify any inconsistent values and convert them to ‘na’ to fill in the gaps with a median value, which will help ensure that our data is accurate and consistent.

```
# Check for missing values:: Before
apply(diabetes_df,2,function(x) sum(is.na(x)))
```

```
##           age           gender  family_diabetes           high_bp
##           0             0             0             0
## physically_active           bmi           smoking           alcohol
##           0             4             0             0
##           sleep        sound_sleep  regular_medicine           junk_food
##           0             0             0             0
##           stress        bp_level           pregnancies           pdiabetes
##           0             0             42             0
##           uration_freq        diabetic
##           0             0
```

Summary:

I noticed some issues with the data in the 'bmi' column of the Excel spreadsheet. Most entries were numbers, but there were a few anomalies like 'half an hr' and one empty field, which resulted in four missing values in the 'bmi' column. I suggest using the median value to replace the missing data to fix this as I did not feel the need to remove the missing values completely. We must also ensure that the entire column has the same data type and handle unexpected data like 'half an hr'. Also, corrected the naming of a column to a suitable name that best represent the data values - 'pregnacies'. Casted the types of sleep, bmi, and pregnancies to numeric.

```
diabetes_df$sleep <- as.numeric(diabetes_df$sleep)
diabetes_df$bmi <- as.numeric(diabetes_df$bmi)
diabetes_df$pregnacies <- as.numeric(diabetes_df$pregnacies)

# Correcting the 'pregnacies' column name to a suitable representation 'pregnacies'
colnames(diabetes_df)[which(names(diabetes_df) == 'pregnacies')] <- 'pregnacies'

# Impute Median for Missing Values in 'pregnacies'
diabetes_df$pregnacies[diabetes_df$pregnacies == ""] <- NA
diabetes_df$pregnacies[is.na(diabetes_df$pregnacies)]<-median(diabetes_df$pregnacies,na.rm=TRUE)

# Impute Median for Missing Values in 'bmi'
diabetes_df$bmi[is.na(diabetes_df$bmi)]<-median(diabetes_df$bmi,na.rm=TRUE)
```

Summary:

Confirming no missing values is present and the naming convention is suitable for analysis.

```
# Check for missing values:: After
apply(diabetes_df,2,function(x) sum(is.na(x)))
```

```
##           age           gender  family_diabetes           high_bp
##           0             0             0             0
## physically_active           bmi           smoking           alcohol
##           0             0             0             0
##           sleep        sound_sleep  regular_medicine           junk_food
##           0             0             0             0
##           stress        bp_level           pregnancies           pdiabetes
##           0             0             0             0
```

```
##      uration_freq      diabetic
##              0              0
```

Creating a new dataframe for Statistical Analysis

```
# diabetes_df will be utilized for EDA
```

```
final_DiabetesDf <- diabetes_df
```

- Transformation

```
# Gender
```

```
# making gender as a factor and One-hot encode 'gender' column
```

```
final_DiabetesDf$gender <- as.factor(final_DiabetesDf$gender)
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(gender = as.integer(gender == "Male"))
```

```
# family history of diabetes
```

```
final_DiabetesDf$family_diabetes <- as.factor(final_DiabetesDf$family_diabetes)
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(family_diabetes = as.integer(family_diabetes == "yes"))
```

```
# high blood pressure
```

```
final_DiabetesDf$high_bp <- as.factor(final_DiabetesDf$high_bp)
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(high_bp = as.integer(high_bp == "yes"))
```

```
# smoking
```

```
final_DiabetesDf$smoking <- as.factor(final_DiabetesDf$smoking)
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(smoking = as.integer(smoking == "yes"))
```

```
# alcohol
```

```
final_DiabetesDf$alcohol <- as.factor(final_DiabetesDf$alcohol)
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(alcohol = as.integer(alcohol == "yes"))
```

```
# regular medicine
```

```
final_DiabetesDf$regular_medicine[final_DiabetesDf$regular_medicine == 'o'] <- 'no'
```

```
final_DiabetesDf$regular_medicine <- as.factor(final_DiabetesDf$regular_medicine)
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(regular_medicine = as.integer(regular_medicine == "yes"))
```

```
# pre diabetes
```

```
final_DiabetesDf$pdabetes <- as.factor(final_DiabetesDf$pdabetes)
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(pdabetes = as.integer(pdabetes == "yes"))
```

```
# diabetes
```

```
# one hot encoding
```

```
final_DiabetesDf <- final_DiabetesDf %>%
```

```
  mutate(
```

```

diabetes_is_yes = as.integer(diabetic == "yes"),
diabetes_is_no = as.integer(diabetic == "no")
) %>%
select(-diabetic)

```

Check for missing values

```
print(apply(final_DiabetesDf, 2, function(x) sum(is.na(x))))
```

```

##          age          gender  family_diabetes          high_bp
##          0           0           0           0
## physically_active          bmi          smoking          alcohol
##          0           0           0           0
##          sleep      sound_sleep  regular_medicine          junk_food
##          0           0           0           0
##          stress      bp_level      pregnancies          pdiabetes
##          0           0           0           0
##          uration_freq  diabetes_is_yes  diabetes_is_no
##          0           0           0

```

```

lapply(final_DiabetesDf, function(col) {
  print(table(col))
})

```

```

## col
##      40-49      50-59 60 or older less than 40
##      164      156      144      488
## col
##  0  1
## 372 580
## col
##  0  1
## 498 454
## col
##  0  1
## 724 228
## col
## less than half an hr more than half an hr          none
##          336          272          132
##      one hr or more
##          212
## col
## 15 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
##  8 16 32 36 48 88 58 76 111 38 66 63 71 28 33 16 16 64 12 12
## 36 38 39 40 42 45
## 20 28  4  4  3  1
## col
##  0  1
## 844 108
## col
##  0  1
## 760 192
## col

```

```

## 4 5 6 7 8 9 10 11
## 32 68 256 264 256 48 24 4
## col
## 0 1 2 3 4 5 6 7 8 9 10 11
## 4 4 44 68 188 156 228 104 120 16 16 4
## col
## 0 1
## 616 336
## col
## always occasionally often very often
## 44 672 184 52
## col
## always not at all sometimes very often
## 88 136 564 164
## col
## high High low Low normal normal
## 211 5 25 3 707 1
## col
## 0 1 2 3 4
## 796 28 64 60 4
## col
## 0 1
## 938 14
## col
## not much quite often
## 664 288
## col
## 0 1
## 686 266
## col
## 0 1
## 268 684

## $age
## col
## 40-49 50-59 60 or older less than 40
## 164 156 144 488
##
## $gender
## col
## 0 1
## 372 580
##
## $family_diabetes
## col
## 0 1
## 498 454
##
## $high_bp
## col
## 0 1
## 724 228
##
## $physically_active

```

```

## col
## less than half an hr more than half an hr          none
##              336              272              132
##      one hr or more
##              212
##
## $bmi
## col
## 15 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
##  8 16 32 36 48 88 58 76 111 38 66 63 71 28 33 16 16 64 12 12
## 36 38 39 40 42 45
## 20 28 4  4  3  1
##
## $smoking
## col
##  0  1
## 844 108
##
## $alcohol
## col
##  0  1
## 760 192
##
## $sleep
## col
##  4  5  6  7  8  9 10 11
## 32 68 256 264 256 48 24  4
##
## $sound_sleep
## col
##  0  1  2  3  4  5  6  7  8  9 10 11
##  4  4  44 68 188 156 228 104 120 16 16  4
##
## $regular_medicine
## col
##  0  1
## 616 336
##
## $junk_food
## col
##      always occasionally      often      very often
##      44      672      184      52
##
## $stress
## col
##      always not at all      sometimes very often
##      88      136      564      164
##
## $bp_level
## col
##      high      High      low      Low      normal normal
##      211      5      25      3      707      1
##
## $pregnancies

```

```
## col
##    0    1    2    3    4
## 796  28  64  60    4
##
## $pdiabetes
## col
##    0    1
## 938  14
##
## $uriation_freq
## col
##      not much quite often
##        664          288
##
## $diabetes_is_yes
## col
##    0    1
## 686 266
##
## $diabetes_is_no
## col
##    0    1
## 268 684
```

Summary:

I converted categorical variables into numerical representations, performed one-hot encoding, and handled some specific cases. For instance, in the case of 'regular_medicine' where the data value was 'o', I used modal imputation to replace it with the most frequently occurring value since it was a categorical variable. In addition, When it comes to identifying outliers, traditional methods like using measures such as the interquartile range (IQR) or standard deviations does not work for categorical variables.

- **Test for Normality**

I will be using ___Anderson Darling test since we have a binary value for target using the ADGofTest library

```
# Anderson-Darling test for presence of diabetes
ad_pres_diabetes <- ad.test(final_DiabetesDf$diabetes_is_yes)
print(ad_pres_diabetes)
```

```
##
## Anderson-Darling GoF Test
##
## data: final_DiabetesDf$diabetes_is_yes
## AD = Inf, p-value = 6.303e-07
## alternative hypothesis: NA
```

```
# Anderson-Darling test for absence of diabetes
ad_absc_diabetes <- ad.test(final_DiabetesDf$diabetes_is_no)
print(ad_absc_diabetes)
```

```
##
```



```
## Anderson-Darling GoF Test
##
## data: final_DiabetesDf$diabetes_is_no
## AD = Inf, p-value = 6.303e-07
## alternative hypothesis: NA
```

Summary:

The Anderson-Darling